

Database Management Systems

CPS 3740

Lecture 9

Computer Science, Kean University

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Outline

- Inner and outer joins
- Chapter 6 hands-on exercises
- Homework and project status
- Midterm 1 next class
 - Midterm coverage
 - Please join the class 5 minutes earlier
 - Cover day 1 – 8 content (exclude SQL injection)
 - SQL injection and day 9 content belong to the scope of the midterm 2 and final exam.

Joins

- **Inner join** - If one row of a joined table is unmatched, row is omitted from result table.
- **Outer join** operations retain rows that do not satisfy the join condition.
 - All the records of one (or both) tables must be displayed.
- Consider following tables:

Branch1

branchNo	bCity
B003	Glasgow
B004	Bristol
B002	London

PropertyForRent1

propertyNo	pCity
PA14	Aberdeen
PL94	London
PG4	Glasgow

Inner Joins

- The (inner) join of these two tables:

SELECT b.*, p.*

FROM Branch1 b, PropertyForRent1 p

WHERE b.bCity = p.pCity;

Branch1

branchNo	bCity
B003	Glasgow
B004	Bristol
B002	London

PropertyForRent1

propertyNo	pCity
PA14	Aberdeen
PL94	London
PG4	Glasgow

branchNo	bCity	propertyNo	pCity
B003	Glasgow	PG4	Glasgow
B002	London	PL94	London

Join two tables

- The (inner) join of these two tables:

```
SELECT b.branchno, city, staffno, salary  
FROM Branch b, Staff s WHERE b.branchno=s.branchno;
```

```
MariaDB [dreamhome]> select * from Branch;
```

branchNo	street	city	postcode
B002	56 Cover Drive	London	NW10 6EU
B003	163 Main Street	Glasgow	G11 9QX
B004	32 Manse Road	Bristol	BS99 1NZ
B005	22 Deer Road	London	SW1 4EH
B007	16 Argyll Street	Aberdeen	AB2 3SU

```
MariaDB [dreamhome]> select * from Staff;
```

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

branchno	city	staffno	salary
B007	Aberdeen	SA9	9000.00
B003	Glasgow	SG14	18000.00
B003	Glasgow	SG37	12000.00
B003	Glasgow	SG5	24000.00
B005	London	SL21	30000.00
B005	London	SL41	9000.00

NATURAL JOIN ($\bowtie$)

- NATURAL JOIN on two relations R and S
 - A binary operator that is written as (R $\bowtie$ S)
 - No join condition specified
 - The result of the natural join is the set of all combinations of tuples in R and S that are equal on their common attribute names.

select * from Branch natural join Staff;

Select * from Branch b, Staff s where s.branchno=b.branchno;

branchNo	street	city	postcode	staffNo	fName	lName	position	sex	DOB	salary
B007	16 Argyll Street	Aberdeen	AB2 3SU	SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00
B003	163 Main Street	Glasgow	G11 9QX	SG14	David	Ford	Supervisor	M	1958-03-24	18000.00
B003	163 Main Street	Glasgow	G11 9QX	SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00
B003	163 Main Street	Glasgow	G11 9QX	SG5	Susan	Brand	Manager	F	1940-06-03	24000.00
B005	22 Deer Road	London	SW1 4EH	SL21	John	White	Manager	M	1945-10-01	30000.00
B005	22 Deer Road	London	SW1 4EH	SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00

Inner Join vs Natural Join

- Inner join:
 - `SELECT * from Staff s, Branch b where s.branchno=b.branchno;`
→ 12 columns, 6 rows
- Natural join:
 - `SELECT * from Staff s natural join Branch b`
→ 11 columns, 6 rows

Outer Joins & Example

- To display rows in the result that do not have matching values in the join column, use Outer join.
- Result table has two rows where cities are same.
- There are no rows corresponding to branches in Bristol and Aberdeen.
- To include unmatched rows in result table, use an Outer join.
- $R \bowtie S$
 - (Left) outer join is join in which tuples from Relation **R** that do not have matching values in common columns of Relation **S** are also included in result relation.

Class exercise - Outer Joins

- List all the branch info and the staff working there. All branch records must be displayed.
- Outer join of these two tables:

```
SELECT b.branchno, city, staffno, salary FROM Branch b  
LEFT JOIN Staff s on b.branchno=s.branchno;
```

```
MariaDB [dreamhome]> select * from Staff;
```

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

```
MariaDB [dreamhome]> select * from Branch;
```

branchNo	street	city	postcode
B002	56 Cover Drive	London	NW10 6EU
B003	163 Main Street	Glasgow	G11 9QX
B004	32 Manse Road	Bristol	BS99 1NZ
B005	22 Deer Road	London	SW1 4EH
B007	16 Argyll Street	Aberdeen	AB2 3SU

branchno	city	staffno	salary
B002	London	NULL	NULL
B003	Glasgow	SG14	18000.00
B003	Glasgow	SG37	12000.00
B003	Glasgow	SG5	24000.00
B004	Bristol	NULL	NULL
B005	London	SL21	30000.00
B005	London	SL41	9000.00
B007	Aberdeen	SA9	9000.00

Example 6.28 Left Outer Join

- Includes those rows of **first (left)** table **unmatched** with rows from second (right) table.
- Columns from second table are filled with **NULLs**.
- List branches and properties that are in same city along with any unmatched branches.

```
SELECT b.*, p.* FROM Branch1 b LEFT JOIN  
PropertyForRent1 p ON b.bCity = p.pCity;
```

Branch1

branchNo	bCity
B003	Glasgow
B004	Bristol
B002	London

PropertyForRent1

propertyNo	pCity
PA14	Aberdeen
PL94	London
PG4	Glasgow

branchNo	bCity	propertyNo	pCity
B003	Glasgow	PG4	Glasgow
B004	Bristol	NULL	NULL
B002	London	PL94	London

Example - Left Outer join

Left outer join: All tuples in the left relation will be retained in the join result, even if it is null or does not have a matching tuple. If it does not have a matching tuple, the corresponding attributes from the right relation in the resultant is kept null.

A

SNO	TOTAL
2008501	211
2008502	137
2008503	210

A ⋈ **B**

SNO	TOTAL	SLNO	ELECTIVE
2008501	211	208703	E 2
2008502	137	207701	E 1

B

SLNO	ELECTIVE	TOTAL
207701	E 1	137
208702	E 2	150
208703	E 2	211

A ⋈ **B**

SNO	TOTAL	SLNO	ELECTIVE
2008501	211	208703	E 2
2008502	137	207701	E 1
2008503	210	null	null

ChatGPT SQL exercises: outer join

- **Step1: Write your queries for the following questions.**
- Given the Staff table (staffNo, fName, lName, position, sex, DOB, salary, branchNo), and Branch table (branchNo, street, city, postcode), write and explain an SQL query to find
 - List all branches and the staff assigned to them. If a branch has no staff, it should still be listed.
 - Find all branches and the number of staff members assigned to each branch. If a branch has no staff, the count should be 0.
 - Find the average salary of staff members for each branch. If a branch has no staff, the average salary should be shown as NULL.

ChatGPT SQL exercises : outer join

Step2: Ask ChatGpt: Copy/paste the above statements into ChatGPT.

Step3: Discussion:

- Who has the correct answers?
- Execute the queries in the database, and verify the results.
- Can you explain your own queries?

Note: ChatGPT is for assistance only. It is not allowed for use during exams. Therefore, you should learn how to solve the problems on your own.

Example 6.29 Right Outer Join

- Right Outer join includes those rows of **second (right)** table that are **unmatched** with rows from first (left) table.
- Columns from first table are filled with **NULLs**.
- List branches and properties in same city and any unmatched properties.

```
SELECT b.*, p.* FROM Branch1 b RIGHT JOIN  
PropertyForRent1 p ON b.bCity = p.pCity;
```

Branch1	
branchNo	bCity
B003	Glasgow
B004	Bristol
B002	London

PropertyForRent1	
propertyNo	pCity
PA14	Aberdeen
PL94	London
PG4	Glasgow

branchNo	bCity	propertyNo	pCity
NULL	NULL	PA14	Aberdeen
B003	Glasgow	PG4	Glasgow
B002	London	PL94	London

Example – Right Outer join

Right Outer Join: All tuples in the right relation will be retained in the join result, even if it is null or does not have a matching tuple.

A

SNO	TOTAL
2008501	211
2008502	137
2008503	210

A ⋈ **B**

SNO	SLNO	ELECTIVE	TOTAL
2008501	208703	E 2	211
2008502	207701	E 1	137

B

SLNO	ELECTIVE	TOTAL
207701	E 1	137
208702	E 2	150
208703	E 2	211

A ⋈_C **B**

SNO	SLNO	ELECTIVE	TOTAL
2008501	208703	E 2	211
2008502	207701	E 1	137
null	207702	E2	150

Example 6.30 Full Outer Join

- Includes rows that are **unmatched** in **both** tables.
 - Unmatched** columns are filled with **NULLs**.
- List branches and properties in same city and any unmatched branches or properties.

```
SELECT b.*, p.* FROM Branch1 b FULL JOIN  
PropertyForRent1 p ON b.bCity = p.pCity;
```

Branch1		PropertyForRent1	
branchNo	bCity	propertyNo	pCity
B003	Glasgow	PA14	Aberdeen
B004	Bristol	PL94	London
B002	London	PG4	Glasgow

branchNo	bCity	propertyNo	pCity
NULL	NULL	PA14	Aberdeen
B003	Glasgow	PG4	Glasgow
B004	Bristol	NULL	NULL
B002	London	PL94	London

Example – Full Outer join

Full Outer Join: All tuples in both left and right relations will be retained in the join result, even if it is null or does not have a matching tuple. Symbol of Full outer join:

A

SNO	TOTAL
2008501	211
2008502	137
2008503	210

A ⋈ **B**

SNO	SLNO	ELECTIVE	TOTAL
2008501	208703	E 2	211
2008502	207701	E 1	137

B

SLNO	ELECTIVE	TOTAL
207701	E 1	137
208702	E 2	150
208703	E 2	211

A ⋈ **B**

SNO	SLNO	ELECTIVE	TOTAL
2008501	208703	E 2	211
2008502	207701	E 1	137
null	207702	E 2	150
2008503	null	null	210

Class exercises - Outer join and subquery

- An outer join may contain a subquery.

```
SELECT ..... FROM T1 RIGHT JOIN  
    (SELECT a1, a2 FROM ... WHERE ...) T2  
    ON T1.a=T2.a1  
);
```

- Exercise on dreamhome database
 - Based on tables:
 - Students_Courses, Year_term
 - Find the year, semester and the number of students took courses. If a year/term has no students taking courses, that year/term should be still in the output.

Class exercises - Outer join and subquery (brainstorm)

- Step 1: Use Students_Courses table to find the number of students took courses for each year/semester.
 - Hint: need to group by
- Step 2: use Year_term to list all the years and semesters and show the numbers of students if there is year/semester matched the step 1 output.

- Result:

year	semester	ct
2013	Spring	NULL
2013	Fall	2
2014	Spring	2
2014	Fall	3
2015	Spring	2
2015	Fall	NULL
2016	Spring	2
2016	Fall	2
2017	Spring	1
2017	Fall	NULL

Class exercises - Outer join and subquery (answer)

- Step 1:

```
select year, semester, count(*) as ct from  
Students_Courses group by year, semester;
```

- Step 2:

- select t1.year, t1.semester, t2.ct from Year_term
t1 **LEFT JOIN** (select year,semester,count(*) as ct
from Students_Courses group by year,semester) t2
ON t1.year=t2.year and t1.semester=t2.semester;

Class exercise – outer join 1

- Use Branch and Staff tables in dreamhome
- List all branchno and staff working there. If a branch has no people, the staff name should be NULL.
 - If Branch at left side, you should use LEFT JOIN.
 - If Branch at right side, you should use RIGHT JOIN.
 - Don't forget the ON condition
- Verify your result
 - Every branchno should be in the output

Class exercise – outer join 1 (Answer)

- List the staff name at every branchno. If the branch has no staff, the name should null
 - select b.branchno, fname, lname from Branch b left join Staff s on b.branchno=s.branchno;

```
MariaDB [dreamhome]> select * from Staff;
```

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

```
select * from Branch;
```

branchNo	street	city	postcode
B002	56 Cover Drive	London	NW10 6EU
B003	163 Main Street	Glasgow	G11 9QX
B004	32 Manse Road	Bristol	BS99 1NZ
B005	22 Deer Road	London	SW1 4EH
B007	16 Argyll Street	Aberdeen	AB2 3SU

branchno	fname	lname
B002	NULL	NULL
B003	David	Ford
B003	Ann	Beech
B003	Susan	Brand
B004	NULL	NULL
B005	John	White
B005	Julie	Lee
B007	Mary	Howe

Class exercise – outer join 2

- Use Courses and Students_Courses, Students tables in dreamhome
- List all courses and the students took the course . If a course never had any student taken, the student information should be NULL.
 - The output should be course id, name, student names, year, semester, grade
- Design your query before you check the answer

Class exercise – outer join 2 (Answer)

Step1: Data analysis

Courses

cid	name	credits
CPS1231	Java1	4
CPS2231	Java2	4
CPS2232	Data Structure	4
CPS3500	Web Programming	3
CPS3740	Database Introductio	3
CPS5920	Database Systems	3
CPS5921	Data Mining	3

Students

sid	first_name	last_name	birthday	major	zipcode
1001	Austin	Huang	1990-01-01	CS	07123
1002	Mary	Smith	1976-02-08	IT	08348
1003	Claudia	Lee	1987-03-21	Math	91384
1004	Andrew	Lin	1947-08-25	Biology	07101
1005	Helen	Wu	1997-01-25	CS	07083
1007	Sarah	Lee	1995-09-15	IT	04101

Students_Courses;

sid	cid	year	semester	grade
1001	CPS2231	2013	Fall	A-
1003	CPS2231	2015	Spring	A
1004	CPS2231	2014	Spring	A-
1001	CPS2232	2016	Fall	C+
1004	CPS2232	2014	Spring	B+
1005	CPS2231	2016	Spring	A
1007	CPS2231	2015	Spring	B
1003	CPS5920	2016	Fall	C
1007	CPS3740	2014	Fall	A
1003	CPS5921	2016	Spring	B
1001	CPS3500	2017	Spring	C+
1003	CPS2231	2014	Fall	F
1004	CPS2232	2014	Fall	C
1004	CPS2231	2013	Fall	C

Class exercise – outer join 2 (Answer)

Step2: write sql

```
select c.cid, c.name, s.first_name, s.last_name, sc.year,  
sc.semester, sc.grade  
from Courses c left join Students_Courses sc on c.cid=sc.cid  
left join Students s on sc.sid=s.sid;
```

Step3: verify result

cid	name	first_name	last_name	year	semester	grade
CPS1231	Java1	NULL	NULL	NULL	NULL	NULL
CPS2231	Java2	Austin	Huang	2013	Fall	A-
CPS2231	Java2	Claudia	Lee	2015	Spring	A
CPS2231	Java2	Andrew	Lin	2014	Spring	A-
CPS2231	Java2	Helen	Wu	2016	Spring	A
CPS2231	Java2	Sarah	Lee	2015	Spring	B
CPS2231	Java2	Claudia	Lee	2014	Fall	F
CPS2231	Java2	Andrew	Lin	2013	Fall	C
CPS2232	Data Structure	Austin	Huang	2016	Fall	C+
CPS2232	Data Structure	Andrew	Lin	2014	Spring	B+
CPS2232	Data Structure	Andrew	Lin	2014	Fall	C
CPS3500	Web Programming	Austin	Huang	2017	Spring	C+
CPS3740	Database Introductio	Sarah	Lee	2014	Fall	A
CPS5920	Database Systems	Claudia	Lee	2016	Fall	C
CPS5921	Data Mining	Claudia	Lee	2016	Spring	B

Class exercise – outer join 3

- Use Tuitions, Courses and Students_Courses tables in dreamhome
- List the number of students and the total tuition income for each year. If a year never had any student attended, both numbers should be 0.
 - The output should be year, number_of_student, income
 - Note: Courses might have different number of credits.
- Design your query before you check the answer

Class exercise – outer join 3 (Answer)

Step1: Data analysis

Tuitions

year	fee_per_credit
2010	100.5
2011	110.5
2012	120.5
2013	130.5
2014	140.5
2015	150.5
2016	160.5
2017	200.5

Courses

cid	name	credits
CPS1231	Java1	4
CPS2231	Java2	4
CPS2232	Data Structure	4
CPS3500	Web Programming	3
CPS3740	Database Introductio	3
CPS5920	Database Systems	3
CPS5921	Data Mining	3

Students_Courses;

sid	cid	year	semester	grade
1001	CPS2231	2013	Fall	A-
1003	CPS2231	2015	Spring	A
1004	CPS2231	2014	Spring	A-
1001	CPS2232	2016	Fall	C+
1004	CPS2232	2014	Spring	B+
1005	CPS2231	2016	Spring	A
1007	CPS2231	2015	Spring	B
1003	CPS5920	2016	Fall	C
1007	CPS3740	2014	Fall	A
1003	CPS5921	2016	Spring	B
1001	CPS3500	2017	Spring	C+
1003	CPS2231	2014	Fall	F
1004	CPS2232	2014	Fall	C
1004	CPS2231	2013	Fall	C

Class exercise – outer join 3 (Answer)

Step2: write sql

```
select t.year, count(sc.sid) as number_of_student,  
IFNULL(sum(fee_per_credit*credits),0) as tuition_income  
from Tuitions t left join Students_Courses sc on t.year=sc.year  
left join Courses c on c.cid=sc.cid group by t.year;
```

Step3: verify result

year	number_of_student	tuition_income
2010	0	0
2011	0	0
2012	0	0
2013	2	1044
2014	5	2669.5
2015	2	1204
2016	4	2247
2017	1	601.5

Review midterm 1 topics

- Please refer to another slide set

More in class SQL exercises

- Work on questions 44 – 77
 - SELECT with aggregations, group by and joins

Leetcode.com SQL exercise

- 175. Combine Two Tables
 - <https://leetcode.com/problems/combine-two-tables/>
- Write a SQL query for a report that provides the following information for each person in the Person table, regardless if there is an address for each of those people:
 - **FirstName, LastName, City, State**

Table: Person

Column Name	Type
PersonId	int
FirstName	varchar
LastName	varchar

PersonId is the primary key column for this table.

Table: Address

Column Name	Type
AddressId	int
PersonId	int
City	varchar
State	varchar

AddressId is the primary key column for this table.

Reading Assignment

- INNER JOIN
- OUTER JOIN